# Surface Chemistry

# Adsorption

The accumulation of molecular species at the surface rather than in the bulk of a solid or liquid is termed adsorption. The molecular species or substance, which concentrates or accumulates at the surface is termed adsorbate and the material on the surface of which the adsorption takes place is called adsorbent (Fig. 1).

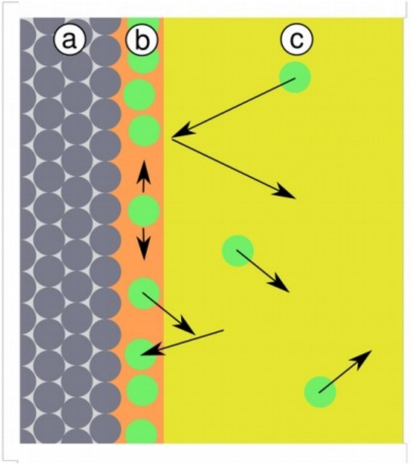



Fig. 1 Adsorption monolayer

# Types of Adsorption

**Physisorption and Chemisorption**

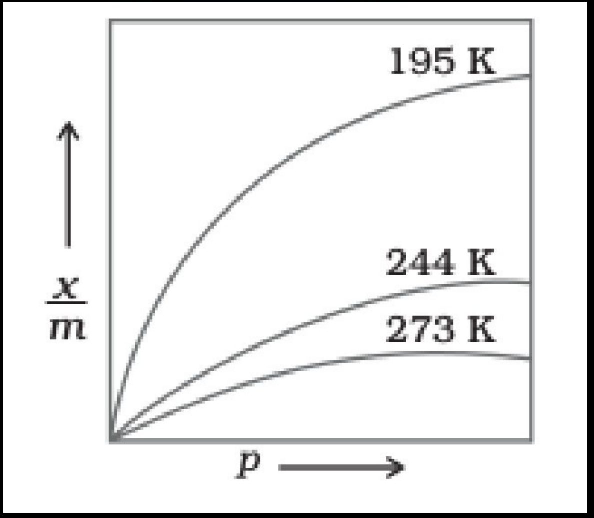
- There are mainly two types of adsorption of gases on solids. If accumulation of gas on the surface of a solid occurs on account of weak van der Waals' forces, the adsorption is termed as **physical adsorption** or **physisorption**.
- When the gas molecules or atoms are held to the solid surface by chemical bonds, the adsorption is termed **chemical adsorption** or **chemisorption**. The chemical bonds may be covalent or ionic in nature. Chemisorption involves a high energy of activation and is, therefore, often referred to as activated adsorption.

| Physisorption | Chemisorption |
|---|---|
| 1. It arises because of van der Waal's forces. | 1. It is caused by chemical bond formation. |
| 2. It is not specific in nature. | 2. It is highly specific in nature. |
| 3. It is reversible in nature. | 3. It is irreversible. |
| 4.It depends on the nature of gas. More liquefiable gases are adsorbed readily. | 4.It also depends on the nature of gas. Gases which can react with the adsorbent show chemisorption. |
| 5.Enthalpy of adsorption is low (20-40 kJ $mol^{-1}$) in this case. | 5.Enthalpy of adsorption high (80-240 kJ $mol^{-1}$) in this case. |
| 6.Low temperature is favourable for adsorption. It decreases with increase of temperature. | 6. High temperature is favourable for adsorption. It increases with the increase of temperature. |
| 7. No appreciable activation energy is needed. | 7. High activation energy is sometimes needed. |
| 8.It depends on the surface area. It increases with an increase of surface area. | 8.It also depends on the surface area. It too increases with an increase of surface area. |
| 9.It results into multimolecular layers on adsorbent surface under high pressure. | 9. It results into unimolecular layer. |

# Freundlich adsorption isotherm

Freundlich, in 1909, gave an empirical relationship between the quantity of gas adsorbed by unit mass of solid adsorbent and pressure at a particular temperature. The relationship can be expressed by the following equation:

$$\boxed{\frac{x}{m} = K.P^{\frac{1}{n}}}$$

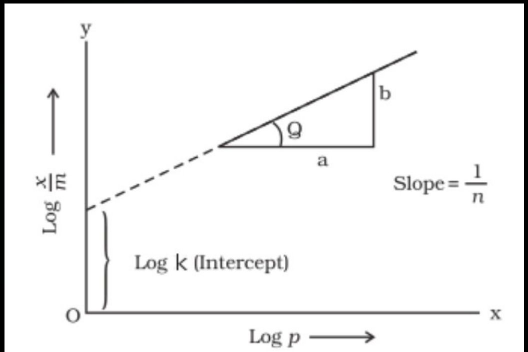



where x is the mass of the gas adsorbed on mass m of the adsorbent at pressure p. whereas k and n are constants which depend on the nature of the adsorbent and the gas at a particular temperature.

# Freundlich adsorption isotherm

Taking logarithm of eq. (1)

$$\text{Log}\,\frac{x}{m} = logK + \frac{1}{n}\,log\text{P}$$




The validity of Freundlich isotherm can be verified by plotting log (x/m) on y-axis (ordinate) and log p on x-axis (abscissa). If it comes to be a straight line, the Freundlich isotherm is valid, otherwise not (Fig. 2). The slope of the straight line gives the value of . The intercept on the y-axis gives the value of log k.

# LANGMUIR ADSORPTION ISOTHERM

- In 1916, Irving Langmuir proposed another Adsorption Isotherm which explained the variation of Adsorption with pressure. Based on his theory, he derived Langmuir Equation which depicted a relationship between the number of active sites of the surface undergoing adsorption and pressure.
- *Assumptions of Langmuir Isotherm*

Langmuir proposed his theory by making following assumptions.

1. Fixed number of vacant or adsorption sites are available on the surface of solid.
2. All the vacant sites are of equal size and shape on the surface of adsorbent.
3. Each site can hold maximum of one gaseous molecule and a constant amount of heat energy is released during this process.
4. Dynamic equilibrium exists between adsorbed gaseous molecules and the free gaseous molecules.
5. Adsorption is monolayer or unilayer

# *Derivation*

Langmuir Equation which depicts a relationship between the number of active sites of the surface undergoing adsorption (i.e. extent of adsorption) and pressure.
To derive Langmuir Equation and new parameter 'θ' is introduced. Let θ the number of sites of the surface which are covered with gaseous molecules. Therefore, the fraction of surface which are unoccupied by gaseous molecules will be (1 – θ).

Now, Rate of forward direction depends upon two factors: Number of sited available on the surface of adsorbent, (1 – θ) and Pressure, P. Therefore, rate of forward reaction is directly proportional to both mentioned factors.

Rate of adsorption a $P(1-\theta)$

Or, rate of adsorption = $K_a P(1-\theta)$

So, Rate of desorption a $\theta$

Or, rate of desorption = $K_d \theta$

At equilibrium, rate of adsorption is equal to rate of desorption, $K_a P (1 - \theta) = K_d \theta$

So, $\boxed{\theta = K_a P / K_d + K_a P}$

- This is known as Langmuir Adsorption Equation.

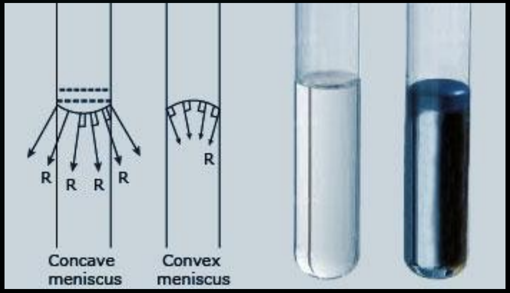
# Surface Tension

- How can we explain

The floating of some objects on the surface of water, even though they are denser than water and the ability of some insects (e.g. water striders) to run on the water surface.

- Definition of surface tension

The attractive force exerted upon the surface molecules of a liquid by the molecules beneath that tends to draw the surface molecules into the bulk of the liquid and makes the liquid assume the shape having the lest surface area.




Surface tension is a property of the surface of a liquid that allows it to resist an external force

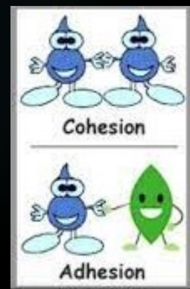
- Effect of surface tension :If we put some water in one tube and some Mercury in another one , the water surface take a convex shape and the Mercury surface take a concave shape

# Cohesive & adhesive force

- Cohesion is the property of like molecules (of the same) to stick to each other due to substance mutual attraction.

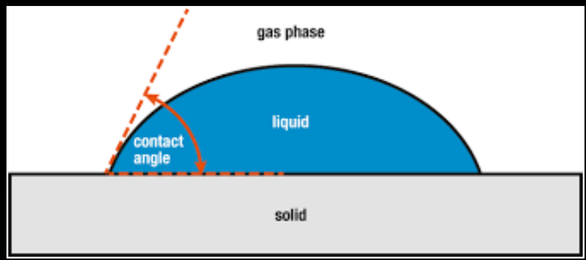
- Adhesion is the property of different molecules or surfaces to cling to each other.

  When the cohesive force of the liquid is stronger than the adhesive force of the liquid to the wall, the liquid concaves down in order to reduce contact with the surface of the all. When the adhesive force of the liquid to the wall is stronger than the cohesive force of the liquid, the liquid is more attracted to the wall than its neighbors, causing the upward concavity.



# Contact Angle

- The contact angle is the angle, conventionally measured through the liquid, where a liquid–vapor interface meets a solid surface.




- Wetting: Wetting is the ability of a liquid to maintain contact with a solid surface, resulting from intermolecular interactions when the two are brought together.
- The degree of wetting (wettability) is determined by a force balance between adhesive and cohesive forces.